

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0913

Sprinkle 138 kV
September 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0913. The Interconnection Customer proposes a 104.4 MW solar photovoltaic facility interconnecting at the Sprinkle 138 kV substation in the Pennsylvania area. The request was received on January 05, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 309 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,304,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SPINDLE 500-345 kV Transformer Bank
Contingency: GREENLAD GAP-SANTEE COPER 345.0 kV
Loading: 109.4% (Rating: 1,049 MVA)
DFAX: 8.8%
Pre-project loading: 108.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SPINDLE 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$24,879,000	5% (\$1,304,000)
	Total allocated to project		\$1,304,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.